Heat and Carno Engine (Supplementary Problems)

1. Compute the maximum possible efficiency of a heat engine operating between the temperature limits of 100 °C and 400 °C.

2. A steam engine operating between a boiler temperature of 220 °C and a condenser temperature of 35.0 °C delivers 8.00 hp. If its efficiency is 30.0 percent of that for a Carnot engine operating between these temperature limits, how many calories are absorbed each second by the boiler? How many calories are exhausted to the condenser each second?

3. A compressor for a jackhammer expands the air in the hammer's cylinder at a constant pressure of 8.6×105 Pa. The increase in the cylinder's volume is 4.05×10^{-4} m^3. During the process, 9.5 J of energy is transferred out of the cylinder as heat.

a. What is the work done by the air?

b. What is the change in the air's internal energy?

c. What type of ideal thermodynamic process does this approximate?

4. A mixture of fuel and air is enclosed in an engine cylinder fitted with a piston. The gas pressure is maintained at 7.07×10^5 Pa as the piston moves slowly inward. If the gas volume decreases by 1.1×10^{-4} m^3 and the internal energy of the gas increases by 62 J, how much energy is added to or removed from the system as heat?

5. Over several cycles, a refrigerator does 1.51×10^4 J of work on the refrigerant. The refrigerant in turn removes 7.55×10^4 J as heat from the air inside the refrigerator.

a. How much energy is transferred as heat to the outside air?

b. What is the net change in the internal energy of the refrigerant?

c. What is the amount of work done on the air inside the refrigerator?

d. What is the net change in the internal energy of the air inside the refrigerator?

6. If a weather balloon in flight gives up 15 J of energy as heat and the gas within it does 13 J of work on the outside air, by how much does its internal energy change?

7. If a steam engine takes in 2.254×10^4 kJ from the boiler and gives up 1.915×10^4 kJ in exhaust during one cycle, what is the engine's efficiency?

8. A test model for an experimental gasoline engine does 45 J of work in one cycle and gives up 31 J as heat. What is the engine's efficiency?

9. A steam engine absorbs 1.98×10^5 J and expels 1.49×10^5 J in each cycle. Assume that all of the remaining energy is used to do work. a. What is the engine's efficiency? b. How much work is done in each cycle?

10. If a gasoline engine has an efficiency of 21 percent and loses 780 J to the cooling system and exhaust during each cycle, how much work is done by the engine?

11. A certain diesel engine performs 372 J of work in each cycle with an efficiency of 33.0 percent. How much energy is transferred from the engine to the exhaust and cooling system as heat?

12. If the energy removed from an engine as heat during one cycle is 6.0×10^2 J, how much energy must be added to the engine during one cycle in order for it to operate at 31 percent efficiency?

13. An engineer claims to have built an engine that takes in 7.5 × 104 J and expels 3.5 × 104 J. a. How much energy can the engine provide by doing work? b. What is the efficiency of the engine? c. Is this efficiency possible? Explain your answer.

14. In one cycle, an engine burning a mixture of air and methanol (methyl alcohol) absorbs 525 J and expels 415 J. What is the engine's efficiency?

15. The energy provided each hour as heat to the turbine in an electric power plant is 9.5×10^{12} J. If 6.5×10^{12} J of energy is exhausted each hour from the engine as heat, what is the efficiency of this heat engine?

16. A heat engine absorbs 850 J of energy per cycle from a high-temperature source. The engine does 3.5×10^2 J of work during each cycle, expelling 5.0×10^2 J as heat. What is the engine's efficiency?

17. A reversible heat engine operates between thermal reservoirs at 900 K and 200 K. If 9000 J are extracted from the hot reservoir, how much work is done by the heat engine?

18. An engine absorbs 3500 J of heat energy each cycle and expels 1200 J.
(a) How much work does it do in one cycle?
(b) If it is a reversible heat engine and the hot reservoir is at 750 K, what is the temperature of the cold reservoir?

19. A gasoline engine absorbs 4000 J of heat and performs 3000 J of mechanical work for each turn of the engine shaft. How much heat is expelled from the engine after 500 turns of the shaft?

20. Consider a reversible heat engine that operates between a hot reservoir at 550 K and a cold reservoir at 150 K. If this engine has a power rating of 750 W, how much heat is expelled to the cold reservoir in 10 min?

21. A reversible heat engine operates between a hot reservoir at 580 K and a cold reservoir of unknown temperature. After 5 min, the engine has absorbed 1000 J of energy from the hot reservoir and expelled 500 J of energy to the cold reservoir. What is the temperature of the cold reservoir?

22. SSM Assume the gasoline engine in your car is a heat engine operating between a hot reservoir at 800 K and a cold reservoir at 280 K and that your engine produces a peak power output of 250 hp (horsepower). If the temperature of the hot reservoir is increased to 900 K, what is the theoretical peak power output?

23. Consider two heat engines: Engine 1 operates between thermal reservoirs at 450 K and 170 K, and engine 2 operates between reservoirs at 50 K and 10 K. Which engine has a greater maximum theoretical efficiency?

24. A reversible heat engine consumes five times more heat QH from its hot reservoir than it expels to its cold reservoir, Q_C. What is the ratio T_H/T_C?

25. Consider a heat engine that operates between thermal reservoirs at 500 K and 250 K. Suppose the engine does 5000 J of work while it expels 1000 J of energy to the cold reservoir. Is this engine allowed by the second law of thermodynamics?

26. A reversible heat pump operating between thermal reservoirs at 25°C and 10°C consumes 500 J of energy each second. How much heat energy is "pumped" into the house each second?

27. A heat pump operates between reservoirs at 293 K (inside the house) and 265 K (outside). It is not a reversible heat pump; it only pumps 75% of the maximum possible heat into the house. How much work must be done to "pump" 6000 J of heat energy into the house?

28. A reversible heat engine operates between thermal reservoirs at 600 K and 150 K. What is the efficiency of this engine?

29. Suppose the heat engine in Problem 28 is operated "in reverse" as a refrigerator. What is its refrigeration efficiency

30. A Carnot engine extracts 2000 J of heat from a hot reservoir and does 1200 J of mechanical work. Suppose the cold reservoir is a very large block of ice at 0°C. What mass of ice would melt during this time?

31. Figure B4.13 represents four successive changes to the state of a gas of constant mass.

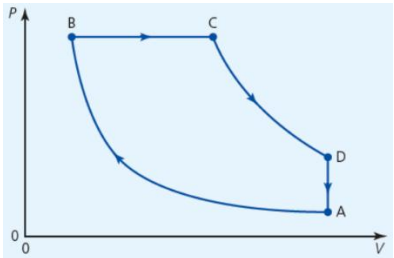


Figure B4.13 Four successive changes to the state of a gas

a. Name the processes shown by:
   i BC ii DA.
b. If CD is an adiabatic process, compare the temperatures at points C and D.
c. If AB is an isothermal process, compare the temperatures at points A and B.

32. Copy and complete Table B4.1 by putting 0, +, –, or ± in each box

Table B4.1

		Q	ΔU	W	ΔP	ΔV	ΔT
isothermal	expansion						
	compression						
adiabatic	expansion						
	compression						
isobaric	expansion						
	compression						
isovolumetric	pressure increase						
	pressure decrease						

33. An ideal monatomic gas expands adiabatically from a volume of 3.13×10^{-3} m^3 to 3.97×10^{-3} m^3.

a. If the original pressure was 2.60 × 105 Pa, determine the final pressure.

b. If the final temperature of the gas was 398 K, what was the starting temperature?

34. Figure B4.15 shows one simplified cycle, ABCA, of a gas in a particular heat engine, during which time 1.3×10^5 J of thermal energy flowed into the gas.

a. Calculate the work done during the process AB.

b. Name the processes AB and BC.

c. Estimate the net useful work done by the gas during the cycle.

d. What is the approximate efficiency of the engine?

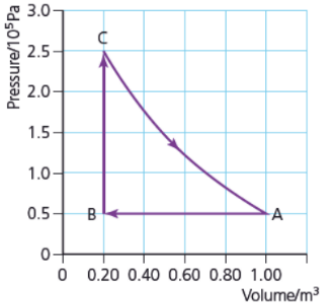


Figure B4.15 An idealized complete cycle

35. Determine the theoretical thermodynamic efficiencies of Carnot cycles operating between temperatures of:
a. 100 °C and 20 °C
b. 500 °C and 100 °C
c. 150 °C and −150 °C.

36. a. With an outlet temperature of 350 K, calculate the inlet temperature needed to achieve a maximum theoretical efficiency of 45% with a Carnot cycle.
b In practice, why will a higher temperature be needed?

37. Sketch a temperature–pressure diagram for the Carnot cycle.

38. Determine the thermodynamic efficiency of a Carnot cycle in which thermal energy is flowing out of the hot reservoir at a rate of 1.26×10^5 W and into the cold reservoir at a rate of 0.79×10^5 W.

39. Figure B4.21 shows the four-stage cycle of a heat engine.

a. Which stage is the compression of the gas?

b. The temperature at A is 320 K. Calculate the amount of gas in moles.

c. Calculate the temperature at point B.

d. Estimate the area ABCD. What does it represent?

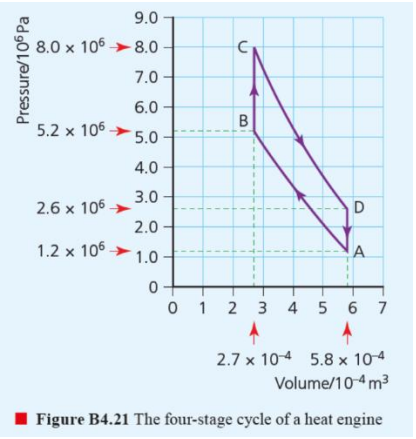


Figure B4.21 The four-stage cycle of a heat engine

40. Using graph paper, make a sketch of the following four consecutive processes in a heat engine. Start your graph at a volume of 20 cm^3 and a pressure of 6.0×10^6 Pa.

a. an isobaric expansion increasing the volume by a factor of five

b. an adiabatic expansion doubling the volume to a pressure of 1.5×10^6 Pa

c. an isovolumetric reduction in pressure to 0.5×10^6 Pa

d. an adiabatic return to its original state.

e. Mark on your graph where work is done on the gas. f Estimate the net work done by the gas during the cycle.

41. A heat engine operates between two reservoirs at temperatures of 20 ˚C and 300 ˚C. What is the maximum efficiency possible for this engine?

42. A steam engine has a boiler that operates at 300 ˚F, and the temperature of the exhaust is 150 ˚F. Find the maximum efficiency of this engine.

43. The energy absorbed by an engine is three times greater than the work it performs.
(a) What is its thermal efficiency?
(b) What fraction of the energy absorbed is expelled to the cold reservoir?

44. A particular engine has a power output of 5.00 kW and an efficiency of 25.0%. If the engine expels 8 000 J of energy in each cycle, find (a) the energy absorbed in each cycle and (b) the time required to complete each cycle.

45. One of the most efficient engines ever built is a coal-fired steam turbine engine in the Ohio River valley, driving an electric generator as it operates between 1 870 ˚C and 430 ˚C.
(a) What is its maximum theoretical efficiency?
(b) Its actual efficiency is 42.0%. How much mechanical power does the engine deliver if it absorbs 1.40×10^5 J of energy each second from the hot reservoir?

46. An engine absorbs 1 700 J from a hot reservoir and expels 1 200 J to a cold reservoir in each cycle. (a) What is the engine's efficiency? (b) How much work is done in each cycle? (c) What is the power output of the engine if each cycle lasts for 0.300 s?

47. A power plant has been proposed that would make use of the temperature gradient in the ocean. The system is to operate between 20.0 ˚C (surface water temperature) and 5.00 ˚C (water temperature at a depth of about 1 km).
(a) What is the maximum efficiency of such a system?
(b) If the useful power output of the plant is 75.0 MW, how much energy is absorbed per hour?
(c) In view of your answer to (a), do you think such a system is worthwhile (considering that there is no charge for fuel)?

48. In one cycle, a heat engine absorbs 500 J from a high temperature reservoir and expels 300 J to a low-temperature reservoir. If the efficiency of this engine is 60% of the efficiency of a Carnot engine, what is the ratio of the low temperature to the high temperature in the Carnot engine?

49. A heat engine operates in a Carnot cycle between 80.0 ˚C and 350 ˚C. It absorbs 21 000 J of energy per cycle from the hot reservoir. The duration of each cycle is 1.00 s.
(a) What is the mechanical power output of this engine? (b) How much energy does it expel in each cycle by heat?

50. A nuclear power plant has an electrical power output of 1 000 MW and operates with an efficiency of 33%. If excess energy is carried away from the plant by a river with a flow rate of 1.0×10^6 kg/s, what is the rise in temperature of the flowing water?

51. A 20.0%-efficient real engine is used to speed up a train from rest to 5.00 m/s. It is known that an ideal (Carnot) engine using the same cold and hot reservoirs would accelerate the same train from rest to a speed of 6.50 m/s using the same amount of fuel. If the engines use air at 300 K as a cold reservoir, find the temperature of the steam serving as the hot reservoir.